Abdullah Bakr

ML Engineer | Computer Vision & Medical Imaging

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SUMMARY

Machine learning engineer focused on computer vision and medical imaging. First-author preprint on Swin Transformer chest-disease classification (112,120 X-rays, 0.8495 mean AUC), a five-stage dental X-ray app deployed live, and deepcsv, an open-source library with 17,000+ PyPI downloads. At Shipd I solved 50+ ML challenges and published 5+ of my own. Kaggle Notebooks Expert.

EXPERIENCE

Machine Learning Engineer (Freelance), Shipd Sep 2026 - Present | Remote

- Solved 50+ machine learning challenges end to end on the Shipd platform.
- Published 5+ original end-to-end ML challenges, each with a deterministic dataset-preparation pipeline and reproducible public/private splits so submissions are scored consistently.

Machine Learning Engineering Intern, FlyRank AI Jul 2026 - Sep 2026 | Remote

- Completed 23 assignments plus a capstone, "Google Search Ranking & Discoverability", accepted by the lead mentor.

SELECTED PROJECTS

Swin Transformer Multi-Label Chest Disease Classification (first-author preprint, Research Square, Sep 2026)

- Classified 14 thoracic diseases from 112,120 chest X-rays (30,805 patients), reaching 0.8495 mean AUC with Swin V2-B, by using patient-level splits to prevent leakage, Focal Loss for class imbalance and Grad-CAM++ for interpretability.

Dental X-Ray AI Application (DENTEX 2023 data)

- Built a five-stage pipeline (quadrant detection, tooth detection and numbering, health status, disease type, severity) from 2 YOLO detectors and 3 Swin classifiers and deployed it as a live Streamlit app.

Two-Stage Skin Cancer Detection Pipeline

- Designed Ultralytics segmentation (melanoma vs. not) feeding a Swin V2-S classifier (6 classes), with custom hair augmentation and explicit tracking of the stage-1 miss rate so first-stage errors stay visible downstream.

Synthetic Image Attribution (ICANN 2026)

- Attributed images to 10 text-to-image models at 95%+ accuracy using a dual-stream EfficientNet-B4 (RGB + SRM noise residuals), test-time augmentation and mixed-precision training.

Drone Detection in the Wild

- Reached 91.8% mAP@50 on 1,000+ images with YOLO26m and a ResNet50 classifier plus custom augmentation.

deepcsv - open-source Python library (PyPI)

- Created a data-preparation library with 17,000+ PyPI downloads that auto-walks directories, converts stringified lists to NumPy, exports Parquet and includes a GradientBoosting mode.

SKILLS

Languages: Python
ML / DL: PyTorch, TensorFlow, Keras, scikit-learn, Swin Transformer, EfficientNet, ResNet, YOLO (Ultralytics), transfer learning
Computer Vision: OpenCV, Albumentations, segmentation, object detection, Grad-CAM / Grad-CAM++, test-time augmentation, mixed precision
Data: Pandas, NumPy, exploratory data analysis, data cleaning and preprocessing, Parquet, patient-level splits
Tools: Git, Jupyter, Kaggle, Streamlit, Ultralytics, AWS AI Practitioner

EDUCATION

B.Sc. Information System Technology, Science Valley Academy, Egypt Sep 2025 - expected Jun 2029 | Grade: Very Good

CERTIFICATIONS, HONORS AND LANGUAGES

Certifications: Google Build with AI (Masr); MaharaTech: Deep Learning for Computer Vision; AWS AI Practitioner Challenge (Udacity); Huawei HCIA-AI V4.0; Anthropic AI Fluency
Honors: Kaggle Notebooks Expert | Favikon #8 Egypt (June 2026) | 14K+ LinkedIn followers writing about applied AI/ML
Languages: Arabic (native), English (professional working)